

# Proofs of eleven congruence conjectures of Hanna from the OEIS, and the resolution of three more

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## Abstract

Many entries of the On-Line Encyclopedia of Integer Sequences contributed by Paul D. Hanna define a power series by a functional equation and conjecture a congruence for its coefficients. We prove eleven such conjectures:

- the modulo-three conjecture of A389472;
- the parity conjectures of A120566, A240998, A273958, A295762, A301933, A338633, A338634 and A388734;
- the modulo-three conjectures of A274479 and A377100.

We also resolve three more:

- for A184894 and A107099 we prove the conjectured vanishing modulo 3 outside the exceptional indices. We also show that the stronger reading, non-vanishing at every exceptional index, is false. It fails at the OEIS index 365 of A184894, and at the coefficient of  $x^{729}$  (OEIS index 364) of A107099;
- for A361047 we prove the intended statement, which the entry states with an index shift.

The proofs reduce the functional equations modulo a prime and use the Frobenius identity, formal derivatives, explicit solutions with an  $x$ -adic uniqueness argument, and Lucas's theorem. Every theorem is stated for every integer power series satisfying the defining equation, and every theorem is formally verified in Lean 4. The proofs were found with the assistance of AI coding agents; their role is described at the end of the paper.

## 1 Introduction

Throughout,  $A(x) = \sum_{n \geq 0} a(n)x^n \in \mathbb{Z}[[x]]$ ,  $A'$  is its formal derivative, and for a series  $h$  with  $h(0) = 0$  we write  $A(h)$  or  $A \circ h$  for the composition  $\sum_n a(n)h^n$ . The results are summarised in Table 1; each conjecture is stated in the corresponding entry [2]. The theorems hold for every integer solution of the defining equation, together with the normalisation of the first terms fixed by the entry where the equation alone does not determine it. The equation and these initial terms determine the sequence by a triangular recursion with integer solutions. There, the new coefficient appears with coefficient 1 or 2. Where it appears with 2, the required divisibility holds: for A274479 by Lemma 1 modulo 2, and for A107099 by writing  $A = x + 2B$ . For the continued fractions, the tails exist as  $x$ -adic limits. Hence the theorems apply to the sequences in the OEIS. These existence arguments are not formalised; the Lean theorems hold for every integer solution.

| Entry   | Defining equation                       | Result                                             |
|---------|-----------------------------------------|----------------------------------------------------|
| A389472 | $A(x^2 + x^3) = x^2(1 + A)$             | $3 \mid a(3n - 1), n > 1$                          |
| A240998 | $A^2 = x + A(x + 2x^2)$                 | $a(n) \text{ odd} \iff n = 2^k \ (n \geq 1)$       |
| A295762 | $A(x - 2A(x^2)) = x + A(x^2)$           | $a(n) \text{ odd} \iff n = 2^k \ (n \geq 1)$       |
| A273958 | $xA + x^2A^2 = C^2, C = x + C^2$        | $a(n) \text{ odd} \iff n = 2 \cdot 4^k - 1$        |
| A301933 | $A = x(1 + 4AA')/(1 + AA')$             | $a(n) \text{ odd} \iff n = 2^k$                    |
| A338633 | $1 = A - x/(A - 2^3x/(A - 3^3x/\dots))$ | $a(n) \text{ odd} \iff n = 2^k \ (n \geq 1)$       |
| A338634 | $1 = A - x/(A - 2^4x/(A - 3^4x/\dots))$ | $a(n) \text{ odd} \iff n = 2^k \ (n \geq 1)$       |
| A388734 | $A = 1 + xA^2 + x^2(1 - x)A^3$          | all $a(n)$ odd                                     |
| A120566 | $A = A \circ A - x A \circ A \circ A$   | all $a(n)$ odd ( $n \geq 1$ )                      |
| A377100 | $A = A(x^3)/A(x^2) + A^2$               | $a(n) \equiv 1 \pmod{3} \ (n \geq 1)$              |
| A274479 | $A^2 = A(x^2/(1 - 2x - 4x^2))$          | $a(n) \equiv 1 \pmod{3} \ (n \geq 1)$              |
| A361047 | $A(x - x^3A'^2) = x$                    | $[x^m]A \equiv [m = 3^k] \pmod{3}$                 |
| A184894 | iterates of $x + x^3$                   | vanishing proved; non-vanishing fails at index 365 |
| A107099 | $A \circ A = x + 4x^3$                  | vanishing proved; non-vanishing fails at $x^{729}$ |

Table 1: Summary. Normalisations and precise statements are given in the theorems below.

**Prior work.** The modulo-two conjecture of A389472 was proved earlier by Perez Fontelles [3] and, independently, in the `trureturing` repository [6]; both leave the modulo-three conjecture open. We found no earlier proof of any of the results in this paper in the OEIS entries and their histories, in the literature, or in the public repositories of proofs of OEIS conjectures that we searched (among them [4, 7]). This reports the coverage of our searches, not a guarantee of priority.

## 2 Tools

Reduction of the coefficients modulo a prime  $p$  is a ring homomorphism  $\mathbb{Z}[[x]] \rightarrow \mathbb{F}_p[[x]]$ ; it commutes with formal differentiation and with composition by series with zero constant term. We write  $F = \overline{A}$  for the reduction and  $f(n)$  for its coefficients.

**Lemma 1** (Frobenius). *For  $F \in \mathbb{F}_p[[x]]$  we have  $F(x)^p = F(x^p)$ .*

*Proof.* The map  $G \mapsto G^p$  is a continuous ring endomorphism of  $\mathbb{F}_p[[x]]$  that fixes  $\mathbb{F}_p$  and sends  $x$  to  $x^p$ .  $\square$

**Lemma 2.** *Let  $K$  be a field, let  $h \in xK[[x]]$  have order at least two, and let  $H \in K[[x]]$  satisfy  $H(h) = H$ . Then  $H$  is constant.*

*Proof.* Put  $B = H - H(0)$ , so  $B(h) = B$ . If  $B \neq 0$ , its order  $d$  is positive, while  $\text{ord } B(h) \geq 2d > d$ .  $\square$

**Lemma 3** (dyadic recursion). *Let  $F \in \mathbb{F}_2[[x]]$  satisfy  $F(0) = 0$  and  $F = x + F^2$ . Then for  $n \geq 1$ ,  $f(n) = 1$  if and only if  $n$  is a power of 2.*

*Proof.* By Lemma 1,  $F = x + F(x^2)$ , so  $f(n) = [n = 1] + [2 \mid n]f(n/2)$  for  $n \geq 1$ . By strong induction,  $f(1) = 1$ ,  $f(n) = 0$  for odd  $n \geq 3$ , and  $f(2m) = f(m)$ .  $\square$

We also use the following observation repeatedly: if  $D \in \mathbb{F}_p[[x]]$ ,  $x^{n_0} \mid D$ , and  $x^n \mid D$  implies  $x^{n+1} \mid D$  for all  $n \geq n_0$ , then  $D = 0$ .

### 3 The modulo-three conjecture of A389472

**Theorem 4.** *If  $A(x^2 + x^3) = x^2(1 + A(x))$ , then  $a(3n - 1) \equiv 0 \pmod{3}$  for every  $n > 1$ .*

*Proof.* Let  $F = \overline{A} \in \mathbb{F}_3[[x]]$  and  $h = x^2 + x^3$ . Then  $F(h) = x^2(1 + F)$  and  $h' = 2x + 3x^2 = -x$ . Differentiating,  $-xF'(h) = -x(1 + F) + x^2F'$ , so  $F'(h) = 1 + F - xF'$  after cancelling  $-x$ . Differentiating again, the terms  $F'$  cancel and  $-xF''(h) = -xF''$ , so  $F''(h) = F''$ . By Lemma 2,  $F''$  is constant. For  $n > 1$  the coefficient of  $x^{3n-3}$  in  $F''$  is therefore zero, and it equals  $(3n - 1)(3n - 2)f(3n - 1) = 2f(3n - 1)$ .  $\square$

### 4 Parity via the dyadic recursion

**Theorem 5.** *In each of the following cases,  $a(n)$  is odd for  $n \geq 1$  if and only if  $n$  is a power of 2:*

1. (A240998)  $A^2 = x + A(x + 2x^2)$ ;
2. (A295762)  $a(0) = 0$  and  $A(x - 2A(x^2)) = x + A(x^2)$ ;
3. (A301933)  $a(0) = 0$  and  $A(1 + AA') = x(1 + 4AA')$ ;
4. (A338633, A338634)  $A$  satisfies the continued fraction  $1 = A - x/(A - 2^e x/(A - 3^e x/(A - \dots)))$  with  $e = 3$ , resp.  $e = 4$ , in the following sense: there are series  $T_0, T_1, \dots$  with constant terms  $\pm 1$  such that  $(A - 1)T_0 = x$  and  $(A - T_k)T_{k+1} = (k + 2)^e x$  for all  $k \geq 0$ .

Moreover (A273958), if  $C = x + C^2$  and  $xA + x^2A^2 = C^2$ , then  $a(n)$  is odd if and only if  $n = 2 \cdot 4^k - 1$  for some  $k \geq 0$ .

*Proof.* (1) Modulo 2,  $(x + 2x^2)^k \equiv x^k$ , so  $F^2 = x + F$ . Then  $G = F - f(0)$  satisfies  $G(0) = 0$  and  $G^2 = F^2 - f(0) = x + G$ , and Lemma 3 applies to  $G$ . (2) Modulo 2 the inner series reduces to  $x$ , so  $F = x + F(x^2) = x + F^2$ . (3) Modulo 2 the equation reads  $F + F^2F' = x$ . In characteristic 2,  $F'' = 0$  because  $n(n - 1)$  is even. Differentiating gives  $F' = 1$ , hence  $F = x + F^2$ . (4) Modulo 2,  $(A - T_0)T_1 = 0$  with  $T_1$  a unit, so  $T_0 = A$  and  $(A - 1)A = x$ . Then  $B = A - 1$  satisfies  $B(0) = 0$  and  $B + B^2 = x$ , so  $B = x + B^2$ , and  $a(n) = b(n)$  for  $n \geq 1$ .

For A273958, constant terms give  $C(0)^2 = 0$ . By Lemma 3,  $c = \overline{C}$  is the indicator of the powers of 2. Put  $B = x\overline{A}$ , so that  $b(n + 1) = f(n)$  and  $B + B^2 = c^2$ , i.e.  $B(x) + B(x^2) = c(x^2)$ . Hence  $b(n) = 0$  for odd  $n$  and  $b(2m) = b(m) + c(m)$ . The powers of 2 split into the disjoint sets  $\{4^j\}$  and  $\{2 \cdot 4^k\}$ , and strong induction gives  $b(n) = 1$  exactly for  $n = 2 \cdot 4^k$ .  $\square$

### 5 Explicit solutions modulo a prime

In the next results we exhibit the reduction  $F$  explicitly and show that it is the only solution by the  $x$ -adic observation of Section 2.

**Theorem 6** (A388734). *If  $A = 1 + xA^2 + x^2(1 - x)A^3$ , then every  $a(n)$  is odd.*

*Proof.* Modulo 2,  $U = 1/(1 - x)$  is a solution:  $1 + (x + x^2)/(1 - x)^2 = 1/(1 - x)$  because  $x + x^2 \equiv x(1 - x)$ . Subtracting the equations for  $F$  and  $U$  gives  $F - U = (F - U)x[F + U + x(1 - x)(F^2 + FU + U^2)]$ , so  $x^n \mid F - U$  implies  $x^{n+1} \mid F - U$ . Hence  $F = U$ .  $\square$

**Theorem 7** (A377100). *If  $a(0) = 0$ ,  $a(1) = 1$  and  $A = A(x^3)/A(x^2) + A^2$ , then  $a(n) \equiv 1 \pmod{3}$  for  $n \geq 1$ .*

*Proof.* Modulo 3 the equation reads  $F(x^2)(F - F^2) = F(x^3)$ . The series  $E = x/(1 - x)$  is a solution: both sides equal  $x^3/(1 - x)^3$ , because  $(1 - x)^3 = 1 - x^3$  and  $1 - 2x \equiv 1 + x$ . For  $D = F - E$  one computes  $E(x^2)D(1 - F - E) = D(x^3) - D(x^2)(F - F^2)$ . If  $x^n \mid D$  with  $n \geq 2$ , the right-hand side is divisible by  $x^{2n+1}$ ; since  $E(x^2)$  is  $x^2$  times a unit and  $1 - F - E$  is a unit,  $x^{2n-1} \mid D$ . As  $x^2 \mid D$ , we get  $F = E$ .  $\square$

**Theorem 8** (A274479). *If  $a(0) = 0$ ,  $a(1) = 1$  and  $A^2 = A(x^2/(1 - 2x - 4x^2))$ , then  $a(n) \equiv 1 \pmod{3}$  for  $n \geq 1$ .*

*Proof.* Modulo 3,  $h = x^2/(1 + x - x^2)$  and  $E = x/(1 - x)$  satisfies  $E(h) = x^2/(1 + x - 2x^2) = x^2/(1 - x)^2 = E^2$ . For  $D = F - E$  we have  $D(F + E) = D(h)$ . If  $x^n \mid D$  then  $x^{2n} \mid D(h)$ , and  $F + E = xW$  with  $W(0) = 2$  a unit, so  $x^{2n-1} \mid D$ . As  $x^2 \mid D$ , we get  $F = E$ .  $\square$

**Theorem 9** (A120566). *If  $a(0) = 0$ ,  $a(1) = 1$  and  $A = A \circ A - xA \circ A \circ A$ , then  $a(n)$  is odd for every  $n \geq 1$ .*

*Proof.* Modulo 2,  $E = x/(1 - x)$  satisfies  $E \circ E = x/(1 - 2x) = x$ , so  $E \circ E - xE \circ E \circ E = x - xE = E$ . Let  $D = F - E$  and suppose  $x^n \mid D$  with  $n \geq 2$ . Since composition preserves congruences modulo  $x^n$  and  $F = x + O(x^2)$ , we have  $D \circ F \equiv D$ ,  $E \circ F - E \circ E = D/((1 - F)(1 - E)) \equiv D$ , and  $x(F \circ F \circ F - E \circ E \circ E) \equiv 0$ , all modulo  $x^{n+1}$ . Since  $F \circ F - E \circ E = D \circ F + (E \circ F - E \circ E)$ , subtracting the equations for  $F$  and  $E$  gives  $D \equiv 2D \equiv 0 \pmod{x^{n+1}}$ . As  $x^2 \mid D$ , we get  $F = E$ .  $\square$

**Theorem 10** (A361047). *If  $A(0) = 0$  and  $A(x - x^3A'^2) = x$ , then  $[x^m]A \equiv 1 \pmod{3}$  when  $m$  is a power of 3, and  $[x^m]A \equiv 0 \pmod{3}$  otherwise.*

*Proof.* Modulo 3 let  $S = \sum_{k \geq 0} x^{3^k}$ . Then  $S' = 1$  and, by Lemma 1,  $S - S^3 = S - S(x^3) = x$ , so  $g \circ S = x$  for  $g = x - x^3$ . Hence  $S \circ g = x$ , i.e.  $S(x - x^3S'^2) = x$ . For  $D = F - S$  with  $x^n \mid D$ , the inner series  $g_F = x - x^3F'^2$  and  $g_S = x - x^3S'^2$  differ by  $x^3D'(F' + S')$ , which is divisible by  $x^{n+2}$ . Therefore  $0 = F(g_F) - S(g_S) = D(g_F) + (S(g_F) - S(g_S)) \equiv D \pmod{x^{n+1}}$ . Hence  $F = S$ .  $\square$

*Remark 11.* The entry A361047 writes  $A = \sum_{n \geq 1} a(n)x^{2n-1}$  and conjectures  $a(n) \equiv 1 \pmod{3}$  exactly for  $n = (3^k - 1)/2$ . With this indexing the statement is contradicted by the data ( $a(2) = 1$ ). It is correct for the index  $i = n - 1$ , since  $x^{2i+1}$  with  $i = (3^k - 1)/2$  is  $x^{3^k}$ . Theorem 10 is this intended statement.

## 6 Additive series modulo three

Call  $G \in \mathbb{F}_3[[x]]$  *additive* if  $[x^m]G = 0$  whenever  $m$  is not a power of 3. By Lemma 1,  $G^{3^j} = G(x^{3^j})$ , so the composition of two additive series is additive.

**Theorem 12** (A184894). *Let  $P_0 = x$ ,  $P_{m+1} = P_m + P_m^3$ , and  $a(m) = [x^{2m-1}]P_m$  for  $m \geq 1$ .*

1. *If  $2m - 1$  is not a power of 3, then  $3 \mid a(m)$ .*
2. *If  $2m - 1 = 3^j$ , then  $a(m) \equiv \binom{m}{j} \pmod{3}$ .*
3.  *$3 \mid a(365)$ , although  $2 \cdot 365 - 1 = 3^6$ .*

*Proof.* Modulo 3,  $P_{m+1} = P_m + P_m(x^3)$ , so the coefficients  $c(m, k) = [x^k]P_m$  satisfy  $c(m+1, k) = c(m, k) + [3 \mid k]c(m, k/3)$  with  $c(0, k) = [k = 1]$ . By induction,  $c(m, k) = 0$  unless  $k$  is a power of 3, and  $c(m, 3^j) = \binom{m}{j}$ . Finally  $\binom{365}{6} \equiv 0 \pmod{3}$  by Lucas's theorem, since  $365 = (111112)_3$  and  $6 = (20)_3$ .  $\square$

The entry conjectures that  $a(m) \equiv 0 \pmod{3}$  except at  $m = (3^n + 1)/2$ . Part (1) proves this statement as written, namely vanishing outside the exceptional indices. Part (3) shows that the stronger reading, that  $a(m)$  is nonzero modulo 3 at every exceptional index, is false; the first failure is at  $n = 6$ . By Lucas's theorem,  $a((3^n + 1)/2) \not\equiv 0$  exactly when every base-3 digit of  $n$  except the last is at most 1.

**Theorem 13** (A107099). *Let  $A(0) = 0$ ,  $[x^1]A = 1$  and  $A \circ A = x + 4x^3$ . Then  $3 \mid [x^n]A$  whenever  $n$  is not a power of 3, and  $3 \nmid [x^{729}]A$ .*

*Proof.* Modulo 3 the equation reads  $F \circ F = x + x^3$ . Suppose  $m$  is not a power of 3 and  $f(j) = 0$  for all non-powers  $j < m$ . Let  $T$  be the truncation of  $F$  below degree  $m$ ; it is additive, so  $[x^m]T \circ T = 0$ . Since  $F = T + f(m)x^m + O(x^{m+1})$  and  $f(1) = 1$ , we get  $[x^m]F \circ F = 2f(m)$ , while  $[x^m](x + x^3) = 0$ ; hence  $f(m) = 0$ . Thus  $F = \sum_k c_k x^{3^k}$  is additive, and comparing coefficients of  $x^{3^k}$  in  $F \circ F = \sum_i c_i F(x^{3^i})$  gives  $c_0 = 1$  and  $2c_k + \sum_{0 < i < k} c_i c_{k-i} = [k = 1]$  for  $k \geq 1$ . This yields  $(c_1, \dots, c_6) = (2, 1, 1, 2, 1, 0)$ .  $\square$

The entry writes  $A = \sum_{n \geq 0} a(n)x^{2n+1}$  and conjectures that  $[x^n]A \equiv 0 \pmod{3}$  except at  $n = 3^k$ . Theorem 13 proves this statement as written. It also shows that the stronger reading, non-vanishing at every power of 3, is false:  $[x^{729}]A = a(364) \equiv 0 \pmod{3}$ . The OEIS b-file of A107099 already shows this zero.

## 7 Verification

**Numerical checks.** For each entry, one of two checks was made. Either a prefix of the sequence was recomputed from the defining equation and compared with the b-file of the entry, or the b-file was substituted into the defining equation and checked modulo two primes. The lengths and moduli vary between entries and are recorded with the verification programs [5]. All checks agree with the theorems. They are cross-checks only; the theorems rest on the proofs and on the Lean verification.

**Formal verification.** All theorems are verified in Lean 4 (version 4.33.1, Mathlib [1] commit 0df444a3) for every integer power series satisfying the stated hypotheses. The Lean files are 0eis389472Mod3 (Theorem 4) and, for the other entries, HannaAnnnnnnn with the A-number of the entry (the file HannaA338633 treats both continued fractions). A separate check confirms that A.subst h denotes the composition  $A(h)$ . For example, Theorem 5 for A273958 reads, in ASCII notation,

```
theorem hanna_a273958 (C A : PowerSeries Int) (hC : C = X + C ^ 2)
  (hA : X * A + X ^ 2 * A ^ 2 = C ^ 2) (n : Nat) :
  Odd (coeff n A) <-> exists k, n + 1 = 2 * 4 ^ k
```

The project was rebuilt from a clean directory. For every theorem, `#print axioms` reports only the standard axioms `propext`, `Classical.choice` and `Quot.sound`; no native evaluation is used. The code and logs are available at [5].

## Tool and computational resource disclosure

This work was carried out between 25 and 26 September 2026 in a workflow directed by the author, using two AI coding agents: OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`) and Anthropic Claude Code (model Claude Opus 5.5). The author chose the research programme and directed the agents. Codex found and formalised the proof of Theorem 4. Claude Code agents searched the OEIS for candidate conjectures, found and

formalised the other proofs, and drafted this text. Each proof was checked by a second agent, and the agents searched the literature for previous proofs. The author is not a professional mathematician. The author has read the paper and, using a side-by-side table, compared the statements of the Lean theorems with the theorems stated here. The correctness of the results rests on the Lean formalisation and on the numerical checks described in the previous section. In addition, each proof was checked by an AI agent other than the one that found it, and the formal statements were compared with the OEIS entries by a second agent. AI systems are not authors of this paper; the author takes full responsibility for its content.

## References

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- [3] A. Perez Fontelles, *Every odd-indexed term of OEIS A389472 after the first is even*, 31 August 2026, paper 1042 in [4], <https://github.com/astrafala/Conjectures/blob/58cf4596af4541652fb191e177ef9b7771388d53/paper-sources/01001-01500/01042-PROOF.tex>.
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